Exam: Artificial Intelligence – Algorithms and Application

Module Exam

Summer 2025 Date: 05.09.2025

TECHNISCHE UNIVERSITÄT DARMSTADT

WIRTSCHAFTS INFORMATIK

Important Information

- Please check your exam copy for completeness.
  It covers **22 pages** (cover sheet included).
- Fill out the cover sheet immediately after receiving the exam.
- Use only the examination paper to solve the tasks. If you do not have enough space, you can receive additional paper during the examination. Additional papers must also be marked with your name and matriculation number.
- Please leave a **correction margin of 3 cm**.
- You have a total of **90 minutes** to complete the exam.
- Except for a **non-programmable calculator, no other aids** are allowed in the exam.

We wish you much success!

Please fill out clearly in block letters.

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Matr. No. ..... Course of Study .....

❑ Master
❑ Diplom

Repeater:
❑ yes ❑ no

Section	Max. Points	Achieved Points
1	35	
2	25	
3	30	
Sum	90	

Exam Review („Klausureinsicht“):

(do not fill out before the review)

I have reviewed the corrected exam:

☐ There are no complaints about the correction.

☐ Complaints about the correction exist (see additional sheet).

Date: ……..

Signature: ……....

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1 Basic Concepts and Algorithms (35 Points)

1.1 Please provide the **definition of artificial intelligence** that has been defined during the **Dartmouth conference**. (1 P)

1.2 Please **name *four other scientific domains of AI*** besides **"Planning and Searching"** and **"Computer Vision"** we discussed in the lecture. (2 P)

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1.3 Please **explain** briefly **why** using the ***elbow criterion*** may **not** be an **effective approach** to **selecting** an **optimal cluster number**. (1 P)

1.4 A **delivery drone** needs to **optimize** its **route** in a **crowded city**.
Please **explain**: **What *algorithm*** would **best suit** this **task**? **Justify** your **answer**. (3 P)

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1.5 Please briefly **explain**: **What** is ***CRISP-DM***? Please also **name** its **distinct *process phases***. (4P)

1.6 Please briefly **explain**: What is the **difference between** the ***Mc Culloch-Pitts Neuron*** and the ***perceptron***? **How** do they **differ related** to **learning**? (3 P)

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1.7 Please **explain** the problem of ***Model Autophagy Disorder (MAD)*** in the **context** of **LLMs/ChatGPT**. (3 P)

1.8 Please briefly **explain** the **difference** between ***random sampling*** and ***random walk*** in search algorithms by comparing both concepts. (2 P)

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1.9 Please **define *supervised*** and ***unsupervised learning.*** Please **explain** the **difference** between the two approaches using an example use case. (3 P)

1.10 Please briefly **explain** how ***CAPTCHAs*** function as a **reverse Turing Test**. (4 P)

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1.11 Please briefly **explain** the **central statements** of the ***Moravec's Paradox***. Please **also provide an example** of your own of Moravec's Paradox in today's Artificial Intelligence. (4 P)

1.12 Please **explain**: **What** tends to **happen** to the **training error** in a **linear model** (e.g., a linear regression) as the **polynomial degree increases**? (1 P)

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1.13 Please define the ***no free lunch theorem*** of AI and **explain why** it is **important**. (4 P)

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2 Classification and Entropy (25 Points)

A **company** is **developing** an ***AI-based email spam classifier***. The **dataset** consists **email examples labeled** as either **Spam** or **Not Spam**, based on different word occurrences. **Table 1** summarizes the **dataset distribution** for the **phrase *"Join me on Whatsapp"*** based on **a qualified sample**.

Table 1

Phrase: "Join me on Whatsapp"	Spam (count)	Not Spam (count)
TRUE	40	10
FALSE	20	30

2.1 Please **name**: What **kind** of ***machine learning problem*** is **this**? (1 P)

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2.2 Please **calculate** the ***entropy*** of the **dataset <u>before</u> any classification based on** the **Spam** and **Not Spam labels** (8 P). Use the following **formular** for **entropy**: $- \Sigma (p * \log_2 p)$

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2.3 Please **calculate** the ***weighted entropy*** **<u>after</u> splitting** the **dataset based on** the **presence** of the suspicious **phrase** *"Join me on Whatsapp"*. To compute the **weighted entropy**, **calculate** the **entropy** for **each dataset** and **weight** it **based** on the **number** of **elements** in **each subset**. (6 P)

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2.4 Please **compute** the ***information gain*** of **splitting** on the **phrase**. Use your values from the previous calculations. Please also **explain**: **Would** this be **a good split** for **classification**? **Why** or **why not**? (6 P)

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2.5 Please **explain**: Does **entropy** as a **criterion** to **build** your ***decision tree*** **guarantee** the **optimal split** in **each scenario**? **Why** or **why not**? (4 P)

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3 Programming with Python (30 Points)

In your current project for a big German sport car company, you plan to investigate the reason for engine breaks. To achieve this, you aim to find correlations between different parts to identify the root cause. During your first analysis, you face the following dataset that you received from the Quality department:

PartID	Height (cm)	Height (inches)	Weight
1	68	26,77	120
2	74	29,13	130
3	76	29,92	141
…	…	…	…

3.1 First, you try to **load the dataset** in your IDE. However, the upload **does not work** and **returns** the following **errors**. Please explain the ***reason for the errors*** and ***how to solve them***. By doing so, please provide code in Python that solves this issue. (1 P)

Python Code

```
connection = sqlite3.connect("failures.db")
cursor = connection.cursor()
sql = "SELECT * FROM failure_report"
cursor.execute(sql)

Traceback (most recent call last):
  File "C:\Users\dominik\.spyder-py3\temp.py", line 7, in <module>
  connection = sqlite3.connect("failures.db")
  NameError: name 'sqlite3' is not defined
```

3.2 In a first step, you plan to **cluster similar parts**. For that purpose, you define the following **`distance()` function** based on the **Euclidian distance**:

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$$d(p,q) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2}$$

Please **implement** this function in **Python**. Write down the function in **code** and **comment** your code to explain it. (4 P)

(**Note**: One should be able to use your implemented distance function with input as in the task 3.4)

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3.3 Please **explain** briefly: **Which *L-norm*** corresponds to the **Euclidean distance**? Please **name one** other **L-norm** by **name**. (2 P)

3.4 Please **compute** the **results** of the following **code** (rounded to three decimal places) (1 P):

```
sim_12_cm = distance([68, 120], [74, 130])
```

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3.5 Please **do the same** for the **following similarities**. (5 P)

(**Note** that, e.g., `sim_23_inches` means that you should compute the distance from the point with *PartID=2* to the point with *PartID=3* based on their respective *Height (inches)* and *Weight*.)

Similarity	Result
`sim_23_inches`	
`sim_13_inches`	
`sim_12_inches`	
`sim_23_cm`	
`sim_13_cm`	

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3.6 **Compare** the **results** of **part 2** using **inches** and **centimeters**. Please **discuss** and **explain** briefly the **results**. **How** do you **handle** such behavior in data preprocessing in general? (2 P)

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3.7 To **handle this problem**, you decide to **write** a **Python notebook**. Unfortunately, some **parts** went **missing**. Please **replace** the **missing parts** in the **code** (marked as <MISSING_#>). (8P)

Python Code

```
# --- 1. Load Vehicle Data ---
Import pandas as <MISSING_1>
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
import numpy as np
import os

csv_filename  = "611_car_data.csv"
try:
    df = pd.read_csv(<MISSING_2>)
except FileNotFoundError:
    print(f"Error: The file '{csv_filename}' was not found. Please ensure it exists.")
    exit()
except Exception as e:
    print(f"An error occurred while loading the data: {e}")
    exit()

X = df.drop('is_broken', axis=1)
y = df['is_broken']


# --- 2. Feature Scaling and Encoding ---
numerical_features = X.select_dtypes(include=np.number).columns.tolist()
categorical_features = X.select_dtypes(include='object').columns.tolist()

numerical_transformer = StandardScaler()
categorical_transformer = OneHotEncoder(handle_unknown='ignore')

preprocessor = ColumnTransformer(
    transformers=[
        ('num', numerical_transformer, numerical_features),
        ('cat', categorical_transformer, categorical_features)
    ])


# --- 3. Split Data into Training and Testing Sets ---
X_train, <MISSING_3>, <MISSING_4>, y_test = train_test_split(X, y, test_size=<MISSING_5>,
random_state=42, stratify=y)


# --- 4. Train a Random Forest Classifier ---
model_pipeline = Pipeline(steps=[('preprocessor', preprocessor),
                                 ('classifier', RandomForestClassifier(n_estimators=100,
random_state=42))])

model_pipeline.fit(X_train, y_train)
```

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Please note: The code continues on the next page.
Please note: This code is an extension of the code on the previous page.

Python Code (continued)

```
# --- 5. Evaluate the Model (Optional but good practice) ---
y_pred = model_pipeline.predict(X_test)
accuracy = accuracy_score(y_test, <MISSING_6>)


# --- 6. Identify Feature Importances ---
random_forest_model = model_pipeline.named_steps['classifier']

ohe_feature_names =
model_pipeline.named_steps['preprocessor'].named_transformers_['cat'].get_feature_names_
out(categorical_features)
all_feature_names = numerical_features + list(ohe_feature_names)

importances = random_forest_model.feature_importances_

feature_importances_df = pd.DataFrame({
    'Feature': all_feature_names,
    'Importance': importances
})


feature_importances_df = feature_importances_df.sort_values(by='Importance',
ascending=False)

print("\nTop Features related to 'is_broken' (Feature Importances):")
print(feature_importances_df)

print("\n--- Analysis Complete ---")
print("The 'Importance' column indicates <MISSING_7>.")

print("Higher values suggest <MISSING_8>.")
```

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Please note: There is more free space available on the next page.

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3.8 From the imported libraries above, please **name** the ***library*** that you typically **use** to **create** a **numerical array** or **perform array operations**. (1P).

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3.9 Please **explain** briefly: What **type** of **machine learning algorithm** is ***RandomForestClassifier*** from sklearn.ensemble, **how** does it **work**, and **what** does the ***n_estimators parameter*** **control**? (4 P).

3.10 Please **explain** briefly **why** ***OneHotEncoder*** from sklearn.preprocessing is **used** for **categorical features** like '*brand*' or '*fuel_type*'. (2P)